

Syllabus for

Academic year 2020/2021 >

TMA976 - Real analysis

Matematisk analys, fortsättning

Syllabus adopted 2019-02-15 by Head of Programme (or corresponding)

Owner: [TKTFY](#)

6,0 Credits

Grading: TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail

Education cycle: First-cycle

Major subject: Mathematics, Engineering Physics

Department: 11 - MATHEMATICAL SCIENCES

Teaching language: Swedish

Application code: 57140

Open for exchange students: No

Only students with the course round in the programme plan

Credit distribution

Module	Sp1	Sp2	Sp3	Sp4	Summer course 	No Sp	Examination dates 
0105 Examination 6,0 c Grading: TH		6,0 c					Fri 15/01-2021 pm J  Thu 08/04-2021 am J, Thu 19/08-2021 am J

In programs

[TKTEM ENGINEERING MATHEMATICS, Year 1 \(compulsory\)](#)

[TKTFY ENGINEERING PHYSICS, Year 1 \(compulsory\)](#)

Examiner:

[Michael Björklund](#)

 [Go to course homepage](#)

Eligibility

General entry requirements for bachelor's level (first cycle)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

The same as for the programme that owns the course.

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Aim

The course provides basic knowledge of the fundamental theories within mathematical physics

Learning outcomes (after completion of the course the student should be able to)

Understand the basic concepts and definitions in mathematical analysis

To be able to prove the basic theorems in mathematical analysis in one variable

To be able to formulate and solve linear/separable differential equations

To be able to expand analytic functions in Taylor series

To be able to analyze the asymptotics of certain sequences (for instance coming from linear difference equations and iterations schemes)

To be able to establish convergence/divergence of series using various tests/criteria.

To be able to establish convergence/divergence of power series.

To apply the notions of pointwise and uniform convergence for function series

To compute limits of functions in several variables

To be able to produce proofs independently.

To be able to solve problems which involve two or more of the above.

Content

Ordinary differential equations: linear equations of the first order, separable equations, linear differential equations of arbitrary order with constant coefficients, systems of equations, some special types such as Euler's differential equation. Mathematical models giving rise to differential equations. Numerical solution of differential equations.

Taylor's formula, computation of limits, l'Hospital's rules.

Difference equations. Sequences, series, power series, convergence criteria, solution of differential equations by means of power series. Uniform convergence of function sequences and function series.

The vector space $\mathbb{R}^n$, polar and spherical coordinates, some topological concepts.

Organisation

Lectures and exercises. Computer exercises with Matlab.

Literature

A. Persson, L.-C. Böiers: Analys i en variabel, Studentlitteratur, Lund.

A. Persson, L.-C. Böiers: Analys i flera variabler, Studentlitteratur, Lund.

Övningar till Analys i en variabel, Matematiska institutionen, Lunds tekniska högskola.

Övningar till Analys i flera variabler, Matematiska institutionen, Lunds tekniska högskola.

F. Eriksson, E. Larsson, G. Wahde: Matematisk analys med tillämpningar, del 3.

OTHER LITERATURE

L. Råde, B. Westergren: BETA - Mathematics Handbook, Studentlitteratur, Lund.

E. Pärt-Enander, A. Sjöberg: Användarhandledning för Matlab 6, Uppsala universitet.

Examination including compulsory elements

A written examination.